

Classifying Quadrilaterals

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A **quadrilateral** is a polygon with four sides.

Parallelogram

A **parallelogram** is a quadrilateral with both pairs of opposite sides parallel.

Rectangle

A **rectangle** is a parallelogram with four right angles.

Rhombus

A **rhombus** is a parallelogram with four congruent sides.

Square

A **square** is a quadrilateral that is both a rectangle and a rhombus.

Trapezoid

A **trapezoid** is a quadrilateral with exactly one pair of parallel sides.

Kite

A **kite** is a quadrilateral with two distinct pairs of adjacent congruent sides.

Classifying Quadrilaterals

Quadrilaterals can be classified using their **sides, angles, parallel sides, and congruent sides.**

Examples

Example 1

A quadrilateral has two pairs of opposite sides parallel and four right angles.

Rectangle

Example 2

A quadrilateral has four congruent sides and four right angles.

Square

Example 3

A quadrilateral has exactly one pair of parallel sides.

Trapezoid

Example 4

A quadrilateral has two pairs of opposite sides parallel and all sides congruent.

Rhombus

Example 5

A quadrilateral has two distinct pairs of adjacent congruent sides.

Kite

Practice Exercises

Practice 1

A quadrilateral has two pairs of opposite sides parallel and four right angles.

Classify the quadrilateral using the most specific name possible.

Practice 2

A quadrilateral has four congruent sides and four right angles.

Classify the quadrilateral using the most specific name possible.

Practice 3

A quadrilateral has exactly one pair of parallel sides.

Classify the quadrilateral using the most specific name possible.

Practice 4

A quadrilateral has two pairs of opposite sides parallel and all sides congruent.

Classify the quadrilateral using the most specific name possible.

Practice 5

A quadrilateral has two distinct pairs of adjacent congruent sides.

Classify the quadrilateral using the most specific name possible.

Practice 6

A quadrilateral has opposite sides parallel and opposite sides congruent.

Classify the quadrilateral using the most specific name possible.

Practice 7

A quadrilateral has side lengths 8, 8, 8, and 8, but its angles are not right angles.

Classify the quadrilateral using the most specific name possible.

Practice 8

A quadrilateral has side lengths 10 and 6 for opposite sides and each angle measures 90° . Classify the quadrilateral using the most specific name possible.

Practice 9

A quadrilateral has only one pair of opposite sides parallel, with bases measuring 12 cm and 18 cm.

Classify the quadrilateral using the most specific name possible.

Practice 10

A quadrilateral has side lengths 5, 5, 9, and 9 arranged as two pairs of adjacent congruent sides.

Classify the quadrilateral using the most specific name possible.

Activities

Activity 1

A quadrilateral has two pairs of opposite sides parallel and four right angles.

Classify the quadrilateral using the most specific name possible.

Activity 2

A quadrilateral has four congruent sides and four right angles.

Classify the quadrilateral using the most specific name possible.

Activity 3

A quadrilateral has exactly one pair of parallel sides.

Classify the quadrilateral using the most specific name possible.

Activity 4

A quadrilateral has two pairs of opposite sides parallel and all sides congruent.

Classify the quadrilateral using the most specific name possible.

Activity 5

A quadrilateral has two distinct pairs of adjacent congruent sides.

Classify the quadrilateral using the most specific name possible.

Activity 6

A quadrilateral has opposite sides parallel and opposite sides congruent.

Classify the quadrilateral using the most specific name possible.

Activity 7

A quadrilateral has side lengths 11, 11, 11, and 11, but its angles are not right angles.

Classify the quadrilateral using the most specific name possible.

Activity 8

A quadrilateral has side lengths 14 and 9 for opposite sides and each angle measures 90° . Classify the quadrilateral using the most specific name possible.

Activity 9

A quadrilateral has only one pair of opposite sides parallel, with bases measuring 15 cm and 22 cm.

Classify the quadrilateral using the most specific name possible.

Activity 10

A quadrilateral has side lengths 7, 7, 12, and 12 arranged as two pairs of adjacent congruent sides.

Classify the quadrilateral using the most specific name possible.

Thank you!